

A new class of catalysts for the glycolysis of PET: Deep eutectic solvent@ZIF-8 composite

Rui Wang, Tianlin Wang, Guangren Yu*, Xiaochun Chen*

Beijing Key Laboratory of Membrane Science & Technology, College of Chemical Engineering, Beijing University of Chemical Technology, Beijing 100029, China

ARTICLE INFO

Article history:

Received 18 August 2020

Revised 25 October 2020

Accepted 10 December 2020

Available online 15 December 2020

Keywords:

PET glycolysis

DES

ZIF-8

ABSTRACT

As an excellent catalyst, DES has been used in PET degradation reaction, but its recovery is currently achieved by evaporating a large amount of solvent, which will undoubtedly consume a lot of energy. In this study, we combined DES and ZIF, and synthesized composites of DES@ZIF-8, in which DES was synthesized using acetamide and different metal salts. Then the effects of experimental parameters on the conversion of PET and the yield of BHET were investigated. Under the optimal conditions: PET(5g), ethylene glycol(25g), and catalyst(0.02g) at 195°C for 25min, PET conversion is 100%, BHET yield is 83.2%. After the reaction, the catalyst can be recovered by simple filtration, the catalyst was recycled for six times with negligible loss of catalytic activity. XRD, FT-IR, SEM and TGA were carried out to characterize the catalyst. In addition, possible mechanism was proposed.

© 2020 Elsevier Ltd. All rights reserved.

1. Introduction

Polyethylene terephthalate (PET) is a thermoplastic polyester, which is widely used in manufacturing food packaging bags, soft drink bottles, fiber, plastic film, and insulation [1,2]. The world consumption of PET has exceeded up to 50 million tons per annum and this number is still growing. Due to very high chemical stability, PET has been accumulated causing serious environmental problems in the world. Therefore, recycling of PET is a hot research topic [3,4].

There are two major methods of recycling for PET, mainly mechanical method and chemical method [5,6]. Chemical method is favored by researchers because it can obtain monomers. Chemical method mainly includes hydrolysis, aminolysis, and alcoholysis [7,8], among which the alcoholysis is especially attracting for its mild reaction conditions and less volatile solvents. The alcoholysis using ethylene glycol as solvent is called glycolysis [9,10]. Glycolysis product is ethylene terephthalate, which can be used as the raw material for re-production of fresh PET. [11].

However, the reaction rate of glycolysis without catalyst is extremely low. Many catalysts such as metal salts, metal oxides, ionic liquids have been developed [12–14]. Although these catalysts have good catalytic effects, they still have disadvantages such as harsh reaction conditions and low monomer yield.

Interestingly, the DES catalyst can greatly promote the glycolysis reaction. DES is prepared through simply mixing a hydrogen bond acceptor (HBA) and a hydrogen bond donor (HBD), which has some novel properties such as negligible vapor pressure, low toxicity and tunability of structure-property [15,16]. The papers about DES used as a catalyst for PET glycolysis have recently been reported [17]. From these published papers, we can find that DES catalyst can improve the rate of glycolysis reaction. For example, Wang et al. used urea/zinc chloride DES as a catalyst, PET is completely degraded after only 30 minutes [18]. The excellent catalytic effect of DES is that it can form hydrogen bond with ethylene glycol, which makes the electronegativity of hydroxyl oxygen in ethylene glycol increase, thus increasing the nucleophilic attack ability of ethylene glycol.

However, DES as liquid catalyst is dissolved in the reaction system. Recycling of DES catalyst can be achieved by evaporating a large amount of solvent under reduced pressure, which will undoubtedly consume a lot of energy. In order to solve the difficulty of liquid catalyst recovery, the catalyst is usually immobilized on the solid material [19].

ZIF-8, the zeolite imidazole ester framework, is a porous crystalline material [20]. Due to its high porosity, large specific surface area, channel regularity and topological diversity, ZIF-8 has been used in catalysis, adsorption separation and electrochemistry [20–22]. Hence, we selected ZIF-8 as a supporter and synthesized DES@ZIF-8 to catalyze PET degradation. To our knowledge, the first time to use DES@ZIF-8 as catalyst for PET glycolysis.

* Corresponding authors.

E-mail addresses: gryu@mail.buct.edu.cn (G. Yu), chenxc@mail.buct.edu.cn (X. Chen).

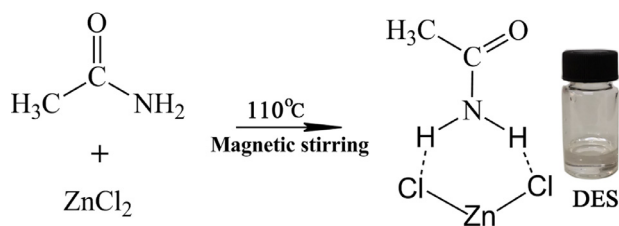


Fig. 1. Synthesis of acetamide/ ZnCl_2 .

In this work, we prepared new DESs where DESs were synthesized using acetamide and four metal salts ZnCl_2 , $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$, $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ and FeCl_3 respectively. Then DESs were immobilized on ZIF-8 separately. The reaction factors are investigated including temperature, catalyst amount, reaction time and EG amount. XRD, FT-IR, SEM and TGA were carried out to study the catalyst structure and thermal stability. This work presents a new class of catalysts for the highly efficient glycolysis of PET.

2. Experimental part

2.1. Materials

PET powder (200 mesh) was purchased from Zhongcheng Plastic Chemical Co., Ltd. Standard sample of BHET(85%) were purchased from Tokyo Chemical Industry Co., Ltd. anhydrous zinc acetate(99%) and 2-methylimidazole(99%) were obtained from Shandong West Asia Chemical Co., Ltd. acetamide(99.5%) was purchased from Sinopharm Chemical Reagent Co., Ltd. acetone(99%), absolute ethanol(99%) and ethylene glycol(99%) were purchased from Beijing Chemical Plant. zinc chloride(98%) was purchased from Tianjin Fuchen Chemical Reagent Factory. The materials were used without any further treatment.

2.2. Instruments and characterization

Catalyst was analyzed by XRD, FT-IR, SEM and TGA. Powder X-ray diffraction analysis was performed at room temperature. It was carried out on a D8 ADVANCE XRD-6000 X-ray diffractometer in reflection mode using $\text{Cu K}\alpha$ radiation ($\lambda = 1.54056\text{\AA}$). The 2θ range from 5° to 50° was scanned with a step size of 0.02° . FT-IR. Fourier transform-infrared (FT-IR) spectra were performed with a Nicolet 6700 Detector spectrometer, using KBr as the blank in the range of $4000\text{--}400\text{ cm}^{-1}$. SEM. The morphologies of the catalyst were characterized using a Hitachi S-4700 scanning electron microscope with an accelerating voltage of 20.0 kV . TGA. Thermogravimetric analyses were recorded on a TGA1/1100 SF STA at a heating rate of 10 K/min under nitrogen atmosphere from 293 to 1073 K .

Main product was analyzed by UPLC-SIR, NMR, MS-ESI and FT-IR. UPLC-SIR analysis of main product was performed using ACQUITY UP-LC BEH C18 (Waters, America). NMR spectra of the main product was recorded on a Bruker AV600 NMR spectrometer operating at 600 MHz in acetone- d_6 solution. The mass spectrum was performed using a ADC Bioscientific G6540B instrument with electrospray ionization (ESI). Fourier transform infrared (FT-IR) measurements were performed using a Nicolet 8700 FTIR instrument using KBr as the blank in the range of $4000\text{--}400\text{ cm}^{-1}$.

2.3. Synthesis of DES

Acetamide was mixed with metal salts at molar ratios of 1:0.4 at 110°C for 4 h, A transparent solution (DES) was obtained. The synthesis diagram is shown in Fig. 1.

2.4. Synthesis of ZIF-8

Typical method of synthesizing ZIF-8 [23,24]: 1.10 g anhydrous zinc acetate and 3.94 g 2-methylimidazole was dissolved in 15 mL deionized water separately and then 2-methylimidazole solution was poured into zinc acetate solution at room temperature gradually. The mixture was stirred for 10 h . Centrifugation was carried out to separate the solid. The obtained solid was washed three times with deionized water and ethanol respectively. The final solid product was kept in oven at 80°C for 24 h for drying. The obtained white powder is ZIF-8.

2.5. Synthesis of DES@ZIF-8 composite

ZIF-8 was mixed with different DESs at mass ratios of 4:1 in absolute ethanol. The solution was vigorously stirred at room temperature for 8 h . Then, the ethanol was distilled off by rotary evaporation to obtain DES@ZIF-8.

2.6. Procedure of PET glycolysis

5 g PET, ethylene glycol and a certain amount of catalyst were mixed in three-necked glass flask equipped with a spherical condenser, a thermometer and magnetic stirrer. The flask was immersed in an oil bath at a specific temperature for the required time.

After the reaction, unreacted PET pellets were quickly separated from the reaction solution and dried at 70°C to constant weight. Then 200 mL boiling water was added to the reaction solution. The solid phase were the catalyst and some oligomers. The catalyst was washed using acetone and dried. 900 mL of cold water was added to the solution, resultantly two phases were obtained and mixture was filtered, the solids were dimer and oligomer; The collected filtrate was concentrated to about 60 mL , and concentrated liquid was stored at 0°C for 12 h for BHET recrystallization and then BHET was filtrated and dried in oven. BHET yield (Y_{BHET}), selectivity (S_{BHET}) and PET conversion (C_{PET}) are calculated as follows:

$$Y_{\text{BHET}} = \frac{W_{\text{BHET}}/MW_{\text{BHET}}}{W_{\text{PET0}}/MW_{\text{PET}}} \times 100\% \quad (1)$$

$$S_{\text{BHET}} = \frac{W_{\text{BHET}}/MW_{\text{BHET}}}{(W_{\text{PET0}} - W_{\text{PET1}})/MW_{\text{PET}}} \times 100\% \quad (2)$$

$$C_{\text{PET}} = \frac{W_{\text{PET0}} - W_{\text{PET1}}}{W_{\text{PET0}}} \times 100\% \quad (3)$$

where W_{BHET} , W_{PET1} and W_{PET0} are the mass (g) of BHET, unreacted PET and original PET, respectively; MW_{BHET} is the molecular weight of BHET, 254 g/mol ; MW_{PET} is a repeating unit molecular weight of PET, 192 g/mol .

3. Results and discussion

3.1. Selection of the catalysts

The catalytic performance of DES, ZIF-8 and DES@ZIF-8 were investigated. The reaction time is determined when PET is completely degraded. The results were shown in Table 1.

Apparently, DES@ZIF-8 presented higher catalytic activity. PET was completely degraded after 25 min and the BHET yield was 83.2% . Surprisingly, the amount of DES@ZIF-8 catalyst was only $0.4\text{ wt\%}$. Compared with pure DES, DES@ZIF-8 had a shorter degradation time. The result indicated that due to the super large specific surface area and porous structure of ZIF-8, the rate of glycolysis reaction was greatly improved. In terms of monomer yield, the catalytic effect of DES@ZIF-8 and pure DES were almost the same,

Table 1

PET glycolysis using DES, ZIF-8, and DES@ZIF-8 where DES is acetamide/ZnCl₂ (PET, 5.0g, 200 mesh; EG, 25.0g; 0.1MPa; 195°C).

Catalyst	Mass ratio of cat./PET	Time (min)	PET conversion (%)	BHET yield (%)	BHET selectivity (%)
DES@ZIF-8	0.4%	25	100.0	83.2	83.2
ZIF-8	0.4%	30	100.0	72.6	72.6
DES	0.4%	100	100.0	76.6	85.6

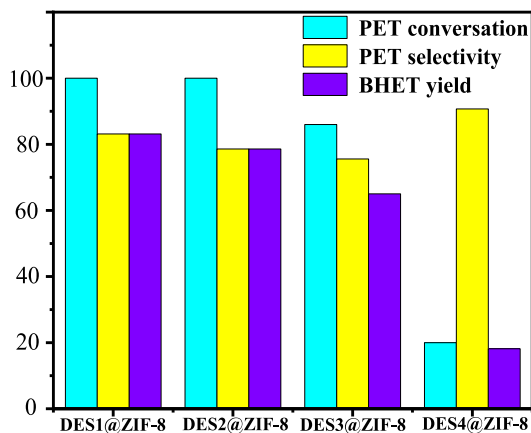


Fig. 2. DES1@ZIF-8, acetamide/ZnCl₂@ZIF-8; DES2@ZIF-8, acetamide/CoCl₂·6H₂O@ZIF-8; DES3@ZIF-8, acetamide/NiCl₂·6H₂O@ZIF-8; DES4@ZIF-8, acetamide/FeCl₃@ZIF-8 (Catalyst, 0.02g; EG, 25 g; PET, 5g; reaction time, 25min; temperature, 195°C).

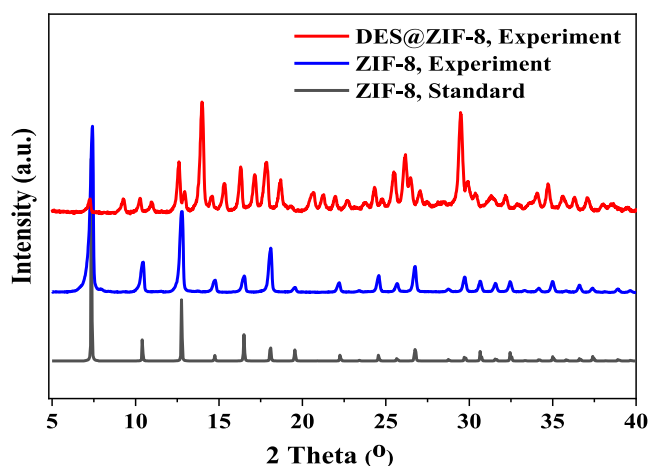


Fig. 3. XRD of ZIF-8 and DES@ZIF-8 where DES is acetamide/ZnCl₂.

indicating that catalytic effect of DES was not greatly affected after being immobilized on ZIF-8. All in all, DES@ZIF-8 had excellent catalytic effects such as short degradation time and high monomer yield.

In order to screen out effective catalyst, four metal salts of ZnCl₂, NiCl₂·6H₂O, CoCl₂·6H₂O and FeCl₃ were reacted with acetamide respectively to form DES. Then DES was immobilized on ZIF-8 to form DES@ZIF-8. The results of PET glycolysis catalyzed by different DES@ZIF-8 were shown in Fig. 2. Acetamide/ZnCl₂@ZIF-8 showed the best catalytic activity in conversion rate and yield under the same reaction conditions. The reason for the difference was speculated to be that the Lewis acidity of Zn is stronger, and it can form a strong coordination effect with the carbonyl oxygen in PET, thereby causing charge transfer, which increases the positive charge on the carbonyl carbon of PET. This role is conducive to the nucleophilic attack of ethylene glycol. We compared DES@ZIF-8 with the reported catalysts (Table 2). It can be

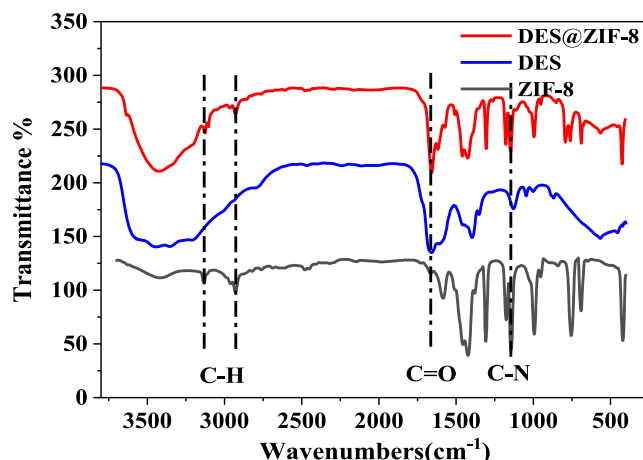


Fig. 4. FT-IR of DES, ZIF-8 and DES@ZIF-8 where DES is acetamide/ZnCl₂.

found that DES1@ZIF-8 has high-efficiency catalysis effect, that is, the higher monomer yield (83.2%) can be achieved using least amount of catalyst (0.4wt%) for shortest time (25 min). Therefore DES1@ZIF-8 has great application potential in PET glycolysis.

3.2. Characterization

Characterization of DES@ZIF-8. The XRD patterns are shown in Fig. 3. The XRD patterns of synthesized ZIF-8 and the standard patterns are consistent. This shows that our method of synthesizing ZIF-8 material is correct and the material has good crystallinity. The XRD of acetamide/ZnCl₂@ZIF-8 show the similar patterns of ZIF-8, indicating that the crystal structure of ZIF-8 is reserved after loading in acetamide/ZnCl₂ DES. The extra peaks can be attributed to the presence of DES. Fig. 4 shows the FT-IR spectrum of acetamide/ZnCl₂@ZIF-8, acetamide/ZnCl₂ and ZIF-8. The strong band at 2933 cm⁻¹ that is assigned to the stretching vibration of C-H in the imidazole ring and the absorption peak near 1655 cm⁻¹ that is assigned to the stretching vibration of a carbonyl group in acetamide indicate acetamide/ZnCl₂ is loaded onto ZIF-8. The SEM picture are shown in Fig. 5. It can be seen that the original ZIF-8 is a hexahedron structure, and has smooth surface and good crystallinity while the surface of DES@ZIF-8 tends to be rough, which indicates ZIF-8 was successfully modified by DES. Fig. 6 shows the TGA curves for DES@ZIF-8 and ZIF-8. At 80–100°C, the 6% mass loss can be attributed to the evaporation of free water on the catalyst surface. At 100–300°C, the slight mass decrease can be attributed to the evaporation of bound water in the sample. In general, DES@ZIF-8 has good thermal stability at 300°C and can be used in actual experiments.

The result of qualitative analysis of the main product could be found in Supporting Information.

3.3. Possible mechanism of PET glycolysis

The published literature about degradation of PET indicated that the hydrogen bonding between catalyst and ethylene glycol is very

Table 2
Comparison of PET degradation with different catalysts.

Catalyst	Mass ratio of cat./PET	Temperature (°C)	Pressure (MPa)	Time(min)	PET conversion (%)	BHET selectivity (%)	BHET yield (%)	Ref.
Zinc acetate	1%	196	0.1	180	100.0	85.6	85.6	[6]
sodium carbonate	0.56%	196	0.1	60	–	80.0	80.0	[25]
ZnMn ₂ O ₄	1%	260	0.5	80	–	–	92.2	[5]
CoMn ₂ O ₄	1%	260	0.5	80	–	–	88.8	
ZnCo ₂ O ₄	1%	260	0.5	80	–	–	80.0	
Mn ₃ O ₄	1%	260	0.5	80	–	–	70.1	
ZnO	1%	260	0.5	80	–	–	61.0	
Co ₃ O ₄	1%	260	0.5	80	–	–	59.7	
β -Zeolite	1%	196	0.1	480	–	–	66.0	[26]
γ -Zeolite	1%	196	0.1	480	–	–	65.0	
γ -	5%	300	1.1	60	100.0	90.0	90.0	[27]
Fe ₂ O ₃	10%	255	1.1	80	100.0	80.0	80.0	
DES1@ZIF-8	0.4%	195	0.1	25	100	83.2	83.2	This work
DES1 @ZIF-8	0.4%	195	0.1	25	100	78.6	78.6	
DES3 @ZIF-8	0.4%	195	0.1	25	86	75	65	
DES4 @ZIF-8	0.4%	195	0.1	25	20	90	18	

Note: DES1@ZIF-8: acetamide/ZnCl₂@ZIF-8; DES2@ZIF-8: acetamide/ CoCl₂·6H₂O @ZIF-8; DES3@ZIF-8: acetamide/ NiCl₂·6H₂O @ZIF-8; DES4@ZIF-8: acetamide/ FeCl₃ @ZIF-8

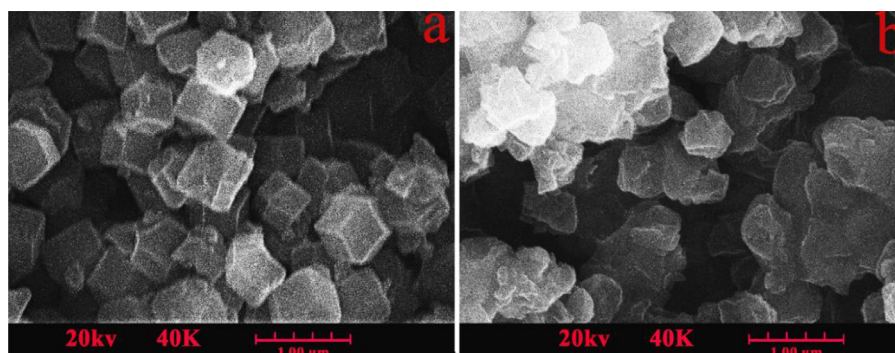


Fig. 5. SEM of ZIF-8 (a) and DES@ZIF-8 (b).

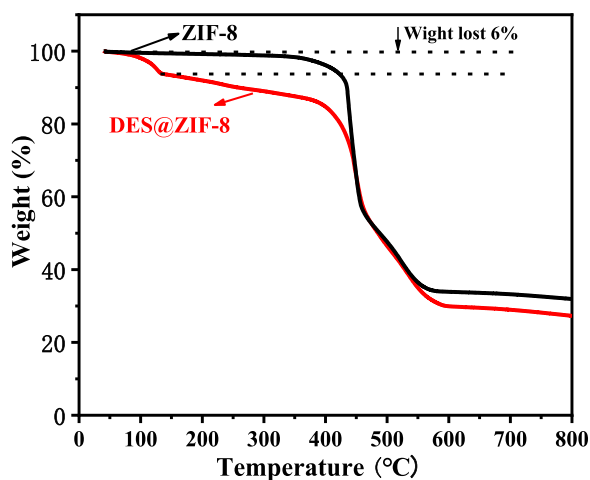


Fig. 6. TGA of ZIF-8 and DES@ZIF-8 where DES is acetamide/ZnCl₂.

important role for degradation of PET [28–30]. Liu et al. used urea derivatives as PET degradation catalysts. they found that the functional group of urea derivatives capable of forming hydrogen bonds with ethylene glycol was an amino group, and the basicity and steric hindrance of the amino group would affect the catalytic effect [17]. For transition metal salt, acting as a Lewis acid catalyst, they mainly coordinate with the carbonyl oxygen of PET [31]. The DES synthesized is acetamide/ZnCl₂, where acetamide has an amino group and can form hydrogen bonds with ethylene glycol and has low steric hindrance because of shorter carbon chain. The

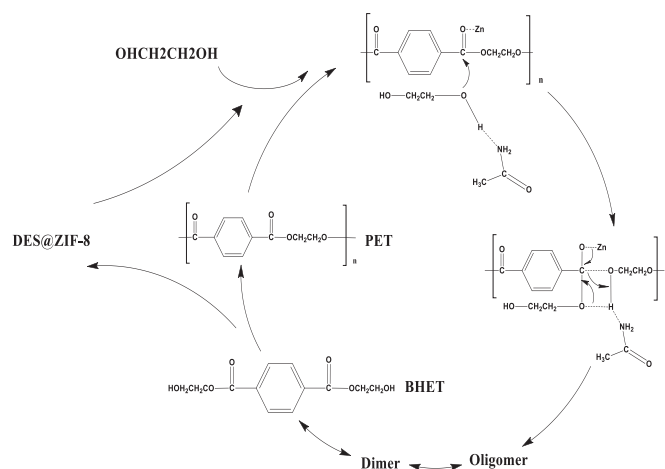


Fig. 7. Mechanism of PET glycolysis

Zn²⁺ in ZnCl₂ can form a strong interaction with the carbonyl oxygen in PET.

Based on the above inferences, the mechanism of PET glycolysis can be explained in Fig. 7. The nitrogen atom in the amino group of acetamide can form a hydrogen bond with the hydroxyl group in ethylene glycol, resulting in enhancing the electronegativity of the hydroxyl oxygen of ethylene glycol. The Zn²⁺ in ZnCl₂ can form coordination with the carbonyl oxygen in PET, leading to increasing the positive charge of carbon to form carbocations, and carbocations are more easily attacked by hydroxyl oxygen of ethylene glycol, so the PET chain is broken.

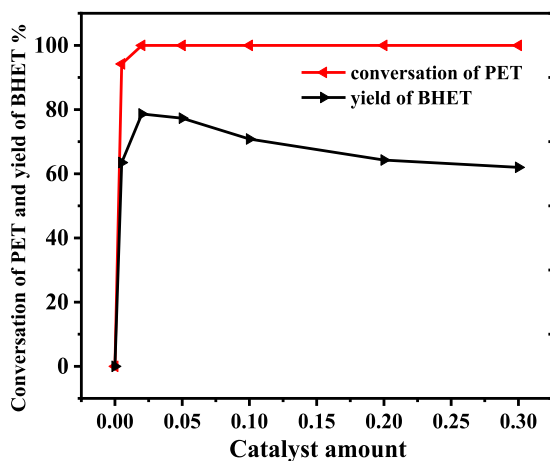


Fig. 8. Effect of catalyst amount of acetamide/ZnCl₂@ZIF-8 on PET conversion and BHET yield (reaction time h; reaction temperature 195°C; EG: 25g; PET: 5g).

It should be noted that there is chemical equilibrium between monomer and dimer, which indicates that if the reaction conditions are not properly controlled, some monomers will be converted into dimers, which is also one of the main reasons for the decline of monomer yield in experiments.

3.4. Investigation of reaction conditions

3.4.1. Catalyst amount

Fig. 8 presents the influence of catalyst amount on the BHET yield and the PET conversion. The result shows that the PET glycolysis did not occur without catalyst. When catalyst amount was 0.02g, the PET conversion reaches 100% and the BHET yield is 78.6%. It can be found that a very small amount of catalyst can greatly promote the reaction rate of PET degradation, while also obtaining high monomer yield. This suggests that DES@ZIF-8 catalyst has a highly efficient catalytic effect and a lower cost. The further increasing catalyst amount, the BHET yield is decreased. After catalyst amount reaching 0.2g, the monomer yield tends to be stable. The possible reason is that there is a chemical balance between BHET and dimers. DES@ZIF-8 catalyst has a very high catalytic effect in a very small amount (0.02g), a lot of BHET were produced in a short time, some BHET converted into by-products (such as dimers) which led to decrease in monomer yield.

3.4.2. Reaction time

Fig. 9 presents the influence of reaction time on the BHET yield and the PET conversion. The 100% PET conversion and 83.2% BHET yield were obtained at 25min, the yield of BHET remains stable between 25–45min. It was observed that further increase in reaction time leads to decrease in BHET yield, e.g. 78.6% at 55min. The reason for the decrease in BHET yield is the chemical balance between BHET and dimer. As the reaction time increases, a part of BHET will be converted into dimer.

3.4.3. Reaction temperature

Fig. 10 displays the influence of reaction temperature on PET-glycolysis. It is showed that temperature at 195°C is sufficient for reaching 100% PET conversion rate and 83.2% BHET yield. When temperature is over 195°C, BHET yield is stable at 83.2%. We can find that the degradation of PET is beneficial with increasing temperature, indicating that the reaction is an endothermic reaction. However, higher temperature might boil EG and cost too much energy, which is not recommended.

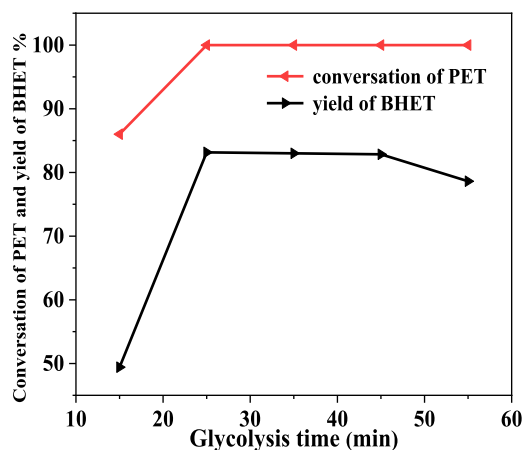


Fig. 9. Effect of reaction time on PET conversion and BHET yield (reaction temperature, 195°C; catalyst amount, 0.02g; EG, 25g; PET, 5g).

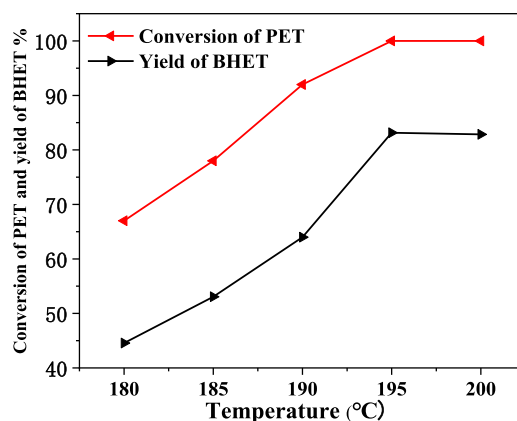


Fig. 10. Effect of reaction temperature on PET conversion and BHET yield (reaction time, 25 min; catalyst amount, 0.02g; EG, 25g; PET, 5g).

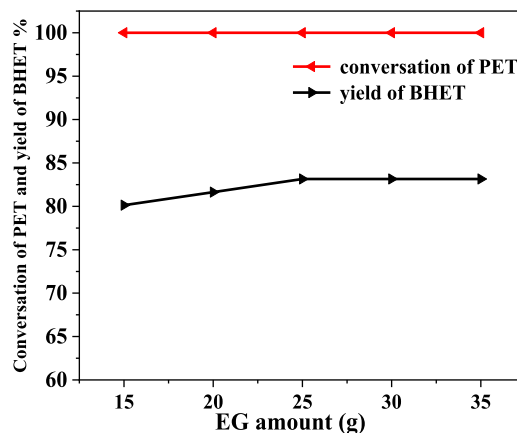


Fig. 11. Effect of the amount of ethylene glycol on PET conversion and BHET yield (reaction time, 25min; reaction temperature, 195°C; catalyst amount, 0.02g; PET, 5g).

3.4.4. EG amount

Fig. 11 presents the influence of EG amount on the BHET yield and the PET conversion. When the EG amount was 25g, the BHET yield was up to 83.2%, above this amount did not change the BHET yield. This phenomenon is attributed to the fact that increase in EG can increase the solubility of PET to promote the degradation reaction, but when the amount of EG exceeds 25g, this effect be-

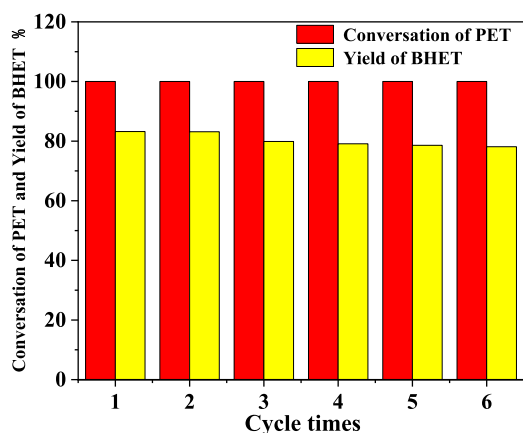


Fig. 12. Recycling of EG and catalyst (reaction temperature, 195°C; catalyst amount, 0.02g; PET, 5g; reaction time, 25min).

comes a little weak to PET degradation, so the BHET yield tends to be stable.

3.5. Recycling of catalyst

When each glycolysis reaction finished, the catalyst can be recovered by simple filtration. After washing and drying, the DES@ZIF-8 were reused for 6 times, and the results are shown in Fig. 12. It can be seen that PET conversion rate is kept 100% and BHET yield slightly decrease, i.e., from 82.3% to 79.5%, which indicates the good stability of this catalyst.

4. Conclusion

We synthesized a new class of catalysts for the glycolysis of PET, that is, the composites of DES@ZIF-8, which proposes a solution for recycling of DES. The factors that influence the catalytic activity such as metal salts species, catalyst amount, reaction time and temperature were investigated. Compared with previously reported catalysts, the composites of DES@ZIF-8 presented much better catalytic performances due to less catalyst amount and less glycolysis time. The synthesized catalyst shows negligible loss of catalytic activity after recycling six times. A catalytic mechanism was proposed finally, in which DES@ZIF-8 containing Zn^{2+} and $-\text{NH}_2$ of acetamide synergistically play the catalytic roles in degradation of PET. To sum up, the composite of DES@ZIF-8 is a new class of catalysts for the glycolysis of PET.

Credit author statement

Rui Wang: experimental operation, Writing- Original draft preparation, Investigation.

Tianlin Wang: experimental operation, Writing and revising- Original draft preparation, Investigation.

Guangren Yu: experimental guidance, Writing- Reviewing and Editing.

Xiaochun Chen: experimental guidance, Writing- Reviewing and Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was financially supported by the National Natural Science Foundation of China (21878010).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.polymdegradstab.2020.109463](https://doi.org/10.1016/j.polymdegradstab.2020.109463).

References

- [1] Q.F. Yue, C.X. Wang, L.N. Zhang, Y. Ni, Y.X. Jin, Glycolysis of poly(ethylene terephthalate) (PET) using basic ionic liquids as catalysts, *Polym. Degrad. Stab.* 96 (4) (2011) 399–403.
- [2] S.B. Wang, C.S. Wang, H.P. Wang, X.L. Chen, S.B. Wang, Sodium titanium tris(glycolate) as a catalyst for the chemical recycling of poly(ethylene terephthalate) via glycolysis and repolycondensation, *Polym. Degrad. Stab.* 114 (2015) 105–114.
- [3] Y.R. Geng, T. Dong, P.T. Fang, Q. Zhou, X.M. Lu, S.J. Zhang, Fast and effective glycolysis of poly(ethylene terephthalate) catalyzed by polyoxometalate, *Polym. Degrad. Stab.* 117 (2015) 30–36.
- [4] M.R. Nabid, Y. Bide, N.B. Fereidouni Etemadi, Maghemite/nitrogen-doped graphene hybrid material as a reusable bifunctional catalyst for glycolysis of polyethylene terephthalate, *Polym. Degrad. Stab.* 144 (2017) 434–441.
- [5] M. Imran, D.H. Kim, W.A. Al-Masry, A. Mahmood, A. Hassan, S. Haider, S.M. Ramay, Manganese-, cobalt-, and zinc-based mixed-oxide spinels as novel catalysts for the chemical recycling of poly(ethylene terephthalate) via glycolysis, *Polym. Degrad. Stab.* 98 (4) (2013) 904–915.
- [6] G.X. Xi, M.X. Lu, C. Sun, Study on depolymerization of waste polyethylene terephthalate into monomer of bis(2-hydroxyethyl terephthalate), *Polym. Degrad. Stab.* 87 (1) (2005) 117–120.
- [7] B. Liu, X.M. Lu, Z.Y. Ju, P. Sun, J.Y. Xin, X.Q. Yao, Q. Zhou, S.J. Zhang, Ultrafast homogeneous glycolysis of waste polyethylene terephthalate via a dissolution-degradation strategy, *Ind. Eng. Chem. Res.* 57 (48) (2018) 16239–16245.
- [8] N. George, T. Kurian, Recent developments in the chemical recycling of post-consumer poly(ethylene terephthalate) waste, *Ind. Eng. Chem. Res.* 53 (37) (2014) 14185–14198.
- [9] H. Wang, R. Yan, Z.X. Li, X.P. Zhang, S.J. Zhang, Fe-containing magnetic ionic liquid as an effective catalyst for the glycolysis of poly(ethylene terephthalate), *Catal. Commun.* 11 (8) (2010) 763–767.
- [10] F.R. Veregue, C.T. Pereira Da Silva, M.P. Moisés, J.G. Meneguín, M.R. Guilherme, P.A. Arroyo, S.L. Favaro, E. Radovanovic, E.M. Giroto, A.W. Rinaldi, Ultrasmall cobalt nanoparticles as a catalyst for PET glycolysis: a green protocol for pure hydroxyethyl terephthalate precipitation without water, *ACS Sustain. Chem. Eng.* 6 (9) (2018) 12017–12024.
- [11] X.Y. Zhou, X.M. Lu, Q. Wang, M.L. Zhu, Z.X. Li, Effective catalysis of poly(ethylene terephthalate) (PET) degradation by metallic acetate ionic liquids, *Pure Appl. Chem.* 84 (3) (2012) 789–801.
- [12] A.M. Al-Sabagh, F.Z. Yehia, A.-M.M.F. Eissa, M.E. Moustafa, G. Eshaq, A.-R.M. Rabie, A.E. Elmetwally, Glycolysis of Poly(ethylene terephthalate) Catalyzed by the Lewis Base Ionic Liquid [Bmim][OAc], *Ind. Eng. Chem. Res.* 53 (48) (2014) 18443–18451.
- [13] F.F. Chen, G.H. Wang, W. Li, F. Yang, Glycolysis of poly(ethylene terephthalate) over Mg–Al mixed oxides catalysts derived from hydrotalcites, *Ind. Eng. Chem. Res.* 52 (2) (2012) 565–571.
- [14] M.L. Zhu, Z.X. Li, Q. Wang, X.Y. Zhou, X.M. Lu, Characterization of solid acid catalysts and their reactivity in the glycolysis of poly(ethylene terephthalate), *Ind. Eng. Chem. Res.* 51 (36) (2012) 11659–11666.
- [15] C. Wu, H.J. Xiao, S.W. Wang, M.S. Tang, Z.L. Tang, W. Xia, W.F. Li, Z. Cao, W.M. He, Natural deep eutectic solvent-catalyzed selenocyanation of activated alkynes via an intermolecular H-bonding activation process, *ACS Sustain. Chem. Eng.* 7 (2) (2019) 2169–2175.
- [16] I. Bodachivskiy, U. Kuzhiumparambil D. B. G. Williams, Catalytic valorization of native biomass in a deep eutectic solvent: a systematic approach toward high-yielding reactions of polysaccharides, *ACS Sustain. Chem. Eng.* 8 (1) (2019) 678–685.
- [17] B. Liu, W.Z. Fu, X.M. Lu, Q. Zhou, S.J. Zhang, Lewis acid–base synergistic catalysis for polyethylene terephthalate degradation by 1,3-dimethylurea/ $\text{Zn}(\text{OAc})_2$ deep eutectic solvent, *ACS Sustain. Chem. Eng.* 7 (3) (2018) 3292–3300.
- [18] Q. Wang, X.Q. Yao, Y.R. Geng, Q. Zhou, X.M. Lu, S.J. Zhang, Deep eutectic solvents as highly active catalysts for the fast and mild glycolysis of poly(ethylene terephthalate)(PET), *Green Chem.* 17 (4) (2015) 2473–2479.
- [19] A.M. Al-Sabagh, F.Z. Yehia, G. Eshaq, A.E. Elmetwally, Ionic liquid-coordinated ferrous acetate complex immobilized on bentonite as a novel separable catalyst for PET glycolysis, *Ind. Eng. Chem. Res.* 54 (50) (2015) 12474–12481.
- [20] G. Kaur, R.K. Rai, D. Tyagi, X. Yao, P.-Z. Li, X.-C. Yang, Y. Zhao, Q. Xu, S.K. Singh, Room-temperature synthesis of bimetallic Co–Zn based zeolitic imidazolate frameworks in water for enhanced CO_2 and H_2 uptakes, *J. Mater. Chem. A* 4 (39) (2016) 14932–14938.

- [21] T.Y. Liang, D. Senthil Raja, K.C. Chin, C.L. Huang, S.A. Sethupathi, L.K. Leong, D.H. Tsai, S.Y. Lu, Bimetallic metal-organic framework-derived hybrid nanostructures as high-performance catalysts for methane dry reforming, *ACS Appl. Mater. Interfaces* 12 (13) (2020) 15183–15193.
- [22] S.H. Kim, Y.J. Lee, D.H. Kim, Y.J. Lee, Bimetallic metal-organic frameworks as efficient cathode catalysts for Li-O₂ batteries, *ACS Appl. Mater. Interfaces* 10 (1) (2018) 660–667.
- [23] M. Garcia-Palacin, J.I. Martinez, L. Paseta, A. Deacon, T. Johnson, M. Malankowska, C.J. Tellez Coronas, Sized-controlled ZIF-8 nanoparticle synthesis from recycled mother liquors: environmental impact assessment, *ACS Sustain. Chem. Eng.* 8 (7) (2020) 2973–2980.
- [24] M. Rana, J. Kim, L. Peng, H. Lim, R. Kaiser, L. Ran, B. Luo, Z. Han, M.S.A. Hossain, X. Lu, I. Gentle, Y. Yamauchi R. Knibbe, Impact of micropores and dopants to mitigate lithium polysulfides shuttle over high surface area of ZIF-8 derived nanoporous carbons, *ACS Appl. Energy Mater.* 3 (6) (2020) 5523–5532.
- [25] R. López-Fonseca, I. Duque-Ingunza, B. De Rivas, L. Flores-Giraldo, J.I. Gutiérrez-Ortiz, Kinetics of catalytic glycolysis of PET wastes with sodium carbonate, *Chem. Eng. J.* 168 (1) (2011) 312–320.
- [26] S.R. Shukla, V.N. Palekar Pingale, Zeolite catalyzed glycolysis of poly(ethylene terephthalate) bottle waste, *J. Appl. Polym. Sci.* 110 (1) (2008) 501–506.
- [27] L. Bartolome, M. Imran, K.G. Lee, A. Sangalang, J.K. Ahn, D.H. Kim, Superparamagnetic γ -Fe₂O₃ nanoparticles as an easily recoverable catalyst for the chemical recycling of PET, *Green Chem.* 16 (1) (2014) 279–286.
- [28] Y.C. Liu, X.Q. Yao, H.Y. Yao, Q. Zhou, J.Y. Xin, X.M. Lu, S.J. Zhang, Degradation of poly(ethylene terephthalate) catalyzed by metal-free choline-based ionic liquids, *Green Chem.* 22 (10) (2020) 3122–3131.
- [29] Q. Wang, Y.R. Geng, X.M. Lu, S.J. Zhang, First-row transition metal-containing ionic liquids as highly active catalysts for the glycolysis of poly(ethylene terephthalate) (PET), *ACS Sustain. Chem. Eng.* 3 (2) (2015) 340–348.
- [30] Q. Wang, X.Q. Yao, S.F. Tang, X.M. Lu, X.P. Zhang, S.J. Zhang, Urea as an efficient and reusable catalyst for the glycolysis of poly(ethylene terephthalate) wastes and the role of hydrogen bond in this process, *Green Chem.* 14 (9) (2012) 2559.
- [31] A.M. Al-Sabagh, F.Z. Yehia, A.M.F. Eissa, M.E. Moustafa, G. Eshaq, A.M. Rabie, A.E. Elmetwally, Cu- and Zn-acetate-containing ionic liquids as catalysts for the glycolysis of poly(ethylene terephthalate), *Polym. Degrad. Stab.* 110 (2014) 364–377.